

Quantification of butyryl CoA:acetate CoA-transferase genes reveals different butyrate production capacity in individuals according to diet and age.



The gastrointestinal microbiota produces short-chain fatty acids, especially butyrate, which affect colonic health, immune function and epigenetic regulation. To assess the effects of nutrition and aging on the production of butyrate, the butyryl-CoA:acetate CoA-transferase gene and population shifts of *Clostridium* clusters IV and XIVa, the main butyrate producers, were analysed. Faecal samples of young healthy omnivores (24 ± 2.5 years), vegetarians (26 ± 5 years) and elderly (86 ± 8 years) omnivores were evaluated. Diet and lifestyle were assessed in questionnaire-based interviews. The elderly had significantly fewer copies of the butyryl-CoA:acetate CoA-transferase gene than young omnivores ($P=0.014$), while vegetarians showed the highest number of copies ($P=0.048$). The thermal denaturation of the butyryl-CoA:acetate CoA-transferase gene variant melting curve related to *Roseburia/Eubacterium rectale* spp. was significantly more variable in the vegetarians than in the elderly. The *Clostridium* cluster XIVa was more abundant in vegetarians ($P=0.049$) and in omnivores ($P<0.01$) than in the elderly group. Gastrointestinal microbiota of the elderly is characterized by decreased butyrate production capacity, reflecting increased risk of degenerative diseases. These results suggest that the butyryl-CoA:acetate CoA-transferase gene is a valuable marker for gastrointestinal microbiota function. © 2011 Federation of European Microbiological Societies. Published by Blackwell Publishing Ltd. All rights reserved.